

Machine Learning for Computational Economics

Spring 2026

Contact Information

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Course Objectives and Learning Outcomes

When applied correctly, “machine learning” tools allow individuals to approximate complicated outcomes in the real world. However, when applied carelessly, these tools generate misleading findings. This course covers supervised learning (both regression and classification), reinforcement learning, and model selection via validation procedures. This course prepares students to apply classical and cutting-edge machine learning techniques to problems in economics.

The course presents a principled approach that adheres to best practices and encourages understanding and transparency.

This class will use Python, a popular high-level computer language, that is being used widely across many fields. “High-level” means it’s less painful than most (the hard work is done by the language), but it’s a serious language with extensive capabilities.

Required Texts and Materials

No textbooks required. There are relevant and useful materials on the QuantEcon website and we will occasionally reference the site but all relevant material is online and open source.

All material will be delivered to students in the form of Jupyter Notebooks.

Exams and Papers

There will be a single exam and a single class project.

The project objectives will be described in a separate document but we have found that an independent (small groups allowed) project where students choose their own topic of interest is one of the most rewarding ways to learn. It allows students to focus in on certain aspects of the course that they find most interesting and explore those more deeply than they otherwise would have been able to.

Grade Policies

There will be 4 categories that make up your grade:

- *Code practice* (30%): These will be homework assignments and we have found that students who complete their homework assignments tend to keep up with the material better and earn “easy” points for their final grade. I will drop your lowest score here. You may work collaboratively with other students on the code practices.
- *Exam* (25%): There will be a single exam. The exam is usually administered as a take-home exam with open note and open internet (if this is abused, we will have to change this policy). This should be individual work and should not be shared with others.
- *Project* (40%): The semester-long project will be the biggest component of your grade. We purposefully give you lots of freedom here so that you can pursue something that you are interested in.
- *Completion* (5%): We expect students to participate in class. If you have questions, please don't be shy about asking them.

All deadlines are a hard-stop and late work will not be accepted.

When something is due “end of day”, you may interpret that to be submit the assignment sometime before 02:30 the following day, i.e. if I say something is due “end of day” Wednesday, then you are permitted to turn it in as late as 2:30 am on Thursday morning (though I recommend getting in the habit of submitting it earlier than that).

Absence Policies

I prefer you attend class and it will be a component of your completion grade. If there is a reason that you cannot attend then please let me know beforehand.

Rice Honor Code

In this course, all students will be held to the standards of the Rice Honor Code, a code that you pledged to honor when you matriculated at this institution. If you are unfamiliar with the details of this code and how it is administered, you should consult the Honor System Handbook at

<http://honor.rice.edu/honor-system-handbook/>. This handbook outlines the University's expectations for the integrity of your academic work, the procedures for resolving alleged violations of those expectations, and the rights and responsibilities of students and faculty members throughout the process.

Disability Resource Center

If you have a documented disability or other condition that may affect academic performance you should: 1) make sure this documentation is on file with the Disability Resource Center (Allen Center, Room 111 / adarice@rice.edu / x5841) to determine the accommodations you need; and 2) talk with me to discuss your accommodation needs.

Syllabus Change Policy

This syllabus is only a guide for the course and is subject to change with advanced notice.

Course Schedule

Recognize that this is a tentative schedule. Depending on class interests and questions, we may spend more time on certain topics and less time on others.

2026-01-16

Syllabus + Course expectations + What is Machine learning + Linear models

2026-01-23

Polynomial regression + overfitting + nonlinear ML

2026-01-30

MLPs + how automatic differentiation works

2026-02-06

PCA + K-Means

2026-02-13

No class

2026-02-20

NLP: Tokenization + naive Bayes

2026-02-27

Gradient descent

2026-03-06

Building and fitting neural networks in PyTorch

2026-03-13

Convolutional neural networks

2026-03-20

No class

2026-03-28

Recurrent neural networks (time-series)

2026-04-03

Recurrent neural networks (NLP)

2026-04-10

Transformers

2026-04-17

Pre-training + Post-training

2026-04-24

Project presentations